Llorenç Balada Gaggioli

Marie Skłodowska-Curie PhD Fellow
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EDUCATION

Czech Technical University in Prague & Université de Toulouse — Prague, Czechia & Toulouse, France
PhD Fellow (Joint doctoral degree) — 10/2024 – Present

- Doctoral Candidate (DC13) in the TENORS network, a Marie Skłodowska-Curie Doctoral Network on tensors.
- Research on quantum optimal control using polynomial optimization and tensor networks.
- Member of the Optimization group at CTU, and the Polynomial Optimization group at LAAS-CNRS.
- Advisors: Dr. Jakub Mareček, Prof. Didier Henrion and Dr. Milan Korda.

The University of Manchester — Manchester, UK
MMath&Phys (Hons) Mathematics and Physics. GPA: 85.95% — 09/2020 – 06/2024

- Relevant courses: Quantum Field Theory, Gravitation, Lie Algebra, Algebraic Topology.
- Master's thesis (1st semester): Study of the Spin-1/2 Kagome Lattice under an External Magnetic Field.
- Master's thesis (2nd semester): Quantum Groups and ZX-calculus.

National University of Singapore — Singapore
Exchange year (3rd year of MMath&Phys). GPA: 4.625/5.0 — 08/2022 – 05/2023

- Relevant courses: Nuclear and Particle Physics, Metric and Topological Spaces.

Salesians de Sarrià — Barcelona, Spain
International Baccalaureate. Grade: 41/45 — 09/2018 – 06/2020

- Courses: Physics HL 7, Mathematics HL 6, Spanish HL 7, Philosophy SL 7, Chemistry SL 6, English SL 6.

RESEARCH & PROFESSIONAL EXPERIENCE

HSBC Quantum Technologies — Singapore & London, UK
Industrial Research Secondment — 09/2026 – 10/2026

- Secondment with HSBC's quantum technologies teams in Singapore (09/2026) and London (10/2026).
- Applying quantum otpimal control methods developed in my PhD to problems in finance.

ICFO – The Institute of Photonic Sciences — Castelldefels, Spain
Summer Fellow — 06/2024 – 09/2024

- Research in the Quantum Information Theory group led by Prof. Antonio Acín.
- Worked on quantum contextuality and Wigner's friend scenarios.

The University of Western Ontario — London, Canada
Mitacs Globalink Research Intern — 05/2023 – 08/2023

- Research in the Quantum Chemistry group under the supervision of Prof. Viktor Staroverov.
- Studied Linearly Independent Product (LIP) basis sets and their use for reconstructing potentials in Density Functional Theory, using tools such as Maple. This work led to a publication in *J. Chem. Phys.*

The University of Manchester — Manchester, UK
Theoretical Physics Research Intern — 06/2022 – 07/2022

- Worked with the Theoretical Physics group on optimizing work extraction from rapidly driven quantum systems, under the supervision of Dr. Harry Miller.
- Used Mathematica for the numerical analysis and visualization of different systems.

Hyperloop Manchester — Manchester, UK
Braking Engineer — 09/2021 – 06/2022

- Studied, developed and optimized electromagnetic braking systems using double-sided Halbach arrays.
- Attended the European Hyperloop Week in Delft (Netherlands) to present our design and compete with the best Hyperloop teams worldwide.

TEACHING & MENTORING

Czech Technical University in Prague, Faculty of Electrical Engineering — Prague, Czechia
Teaching Assistant – Quantum Computing — Winter semester 2025/26

- Teaching duties within the PhD programme for a master's-level course on advanced optimization methods.
- Led tutorial sessions, prepared exercises and supported students with assignments.

The University of Manchester — Manchester, UK
Peer Assisted Study Sessions (PASS) Mentor — 09/2021 – 06/2023

- Supported first-year students by answering their questions and sharing experience from previous years.
- Co-organized and designed the sessions with another PASS leader.

PUBLICATIONS

Journal articles

1. Rufus Lawrence, Aleš Wodecki, Johannes Aspman, **Llorenç Balada Gaggioli**, Jakub Mareček. *Geometric quantum control and the random Schrödinger equation.* Physical Review Research **8**, 013150 (2026).
2. **Llorenç Balada Gaggioli**, Jakub Mareček. *Time evolution of controlled many-body quantum systems with matrix product operators.* Physical Review A **112**, 062612 (2025).
3. Denys I. Bondar, **Llorenç Balada Gaggioli**, Georgios Korpas, Jakub Mareček, Jiri Vala, Kurt Jacobs. *Globally optimal control of quantum dynamics.* Physical Review Research **7**, 043202 (2025).
4. Georgii N. Sizov, Vincent Lazeran, **Llorenç Balada Gaggioli**, Viktor N. Staroverov. *Maximization of linear independence of basis function products.* The Journal of Chemical Physics **160**, 234106 (2024).

Preprints

5. **Llorenç Balada Gaggioli**. *Time-optimal robust quantum control for partially known errors* (2026).
6. Younes Naceur, **Llorenç Balada Gaggioli**. *Reachability and optimal-time certificates for quantum control* (2026).
7. **Llorenç Balada Gaggioli**, Didier Henrion, Milan Korda. *Composition and tensor train structure in polynomial optimization* (2026).
8. **Llorenç Balada Gaggioli**. *Low-rank geometry of two-qubit gates* (2026).
9. **Llorenç Balada Gaggioli**, Didier Henrion, Milan Korda. *Global optimization of low-rank polynomials* (2025).
10. **Llorenç Balada Gaggioli**, Denys I. Bondar, Jiri Vala, Roman Ovsiannikov, Jakub Mareček. *Unitary gate synthesis via polynomial optimization* (2025).

Open-source code for most papers is available at github.com/llorebaga.

Contributed talks

- SIAM Conference on Optimization (OP26), Edinburgh, UK — 06/2026
- DO Workshop, Carcassonne, France — 06/2026
- Optimization Workshop, Spálené Poříčí, Czechia — 11/2025
- Quantum Optimization Workshop, Brdy, Czechia — 11/2024

OTHER EXPERIENCE

Faculty of Science and Engineering, The University of Manchester — Manchester, UK
Student Ambassador — 01/2021 – 01/2023

- Mentored 50 offer-holders by answering their questions and explaining university life.
- Revised and corrected the subtitles of the Faculty's lectures and talks.

Fundació Burriac — Cabrera de Mar, Spain
Book co-author: "Oficis i comerços en la memòria de Cabrera de Mar" — 10/2018 – 06/2020

- Interviewed more than 50 people from Cabrera de Mar, a small town near Barcelona.
- Investigated how jobs in the town evolved during the 20th century.

CosmoCaixa Barcelona — Barcelona, Spain
Museum Guide — 09/2018 – 12/2018

- Explained conceptually challenging museum modules to more than 50 visitors per day (3 h shifts) of a wide range of ages and nationalities.
- Guided visitors through the museum and answered specific scientific questions.

ACHIEVEMENTS & AWARDS

- **Marie Skłodowska-Curie Actions Doctoral Fellowship** (2024): Funds my PhD within the TENORS network (HORIZON-MSCA-2022-DN-JD, grant no. 101120296).
- **ICFO Summer Fellowship** (2024): Stipend to do research in the Quantum Information Theory group at ICFO – Institut de Ciències Fotòniques.
- **Mitacs Globalink Research Internship** (2023): Competitive award for a funded research internship at a Canadian university.
- **DUO Singapore Exchange Fellowship** (2022): Merit-based award from the Singapore Ministry of Foreign Affairs.
- **Padre Damián IV Excellence Award** (2018): For my result in an exam taken by students across Barcelona.
- **AMPA Research Project Award** (2018): For the historical research project "Sant Feliu".

SKILLS & LANGUAGES

Programming: Julia, Python, Mathematica, Maple.
Languages: Catalan (native), Spanish (native), English (fluent).

References available on request.